Intakes of meat, fish, poultry, and eggs and risk of prostate cancer progression

Background: Processed meat and fish have been shown to be associated with the risk of advanced prostate cancer, but few studies have examined diet after prostate cancer diagnosis and risk of its progression. Objective: We examined the association between postdiagnostic consumption of processed and unprocessed red meat, fish, poultry, and eggs and the risk of prostate cancer recurrence or progression. Design: We conducted a prospective study in 1294 men with prostate cancer, without recurrence or progression as of 2004–2005, who were participating in the Cancer of the Prostate Strategic Urologic Research Endeavor and who were followed for an average of 2 y. Results: We observed 127 events (prostate cancer death or metastases, elevated prostate-specific antigen concentration, or secondary treatment) during 2610 person-years. Intakes of processed and unprocessed red meat, fish, total poultry, and skinless poultry were not associated with prostate cancer recurrence or progression. Greater consumption of eggs and poultry with skin was associated with 2-fold increases in risk in a comparison of extreme quantiles: eggs [hazard ratio (HR): 2.02; 95% CI: 1.10, 3.72; P for trend = 0.05] and poultry with skin (HR: 2.26; 95% CI: 1.36, 3.76; P for trend = 0.003). An interaction was observed between prognostic risk at diagnosis and poultry. Men with high prognostic risk and a high poultry intake had a 4-fold increased risk of recurrence or progression compared with men with low/intermediate prognostic risk and a low poultry intake (P for interaction = 0.003). Conclusions: Our results suggest that the postdiagnostic consumption of processed or unprocessed red meat, fish, or skinless poultry is not associated with prostate cancer recurrence or progression, whereas consumption of eggs and poultry with skin may increase the risk.